

Campus Services Portal - Frontend Technical Documentation

Executive Overview

The **Campus Services Portal** frontend is an enterprise single-page application built on **Angular 19**. It features a 100% standalone component architecture, fine-grained Angular Signals reactivity, lazy-loaded routing modules, typed HTTP services, and strict adherence to university business requirements.

Part 1: Component Inventory (104 Components)

The table below catalogs every component within the frontend workspace, detailing its CSS selector, file path, inputs, outputs, and functional role.

COMPONENT CLASS	SELECTOR	CATEGORY	INPUTS	OUTPUTS	RELATIVE FILE PATH
AdminDashboardPageComponent	app-admin-dashboard-page	Admin Feature	None	None	Frontend/src/app/features/admin/dashboard/pages/admin-dashboard-page.component.ts
AdminManagementPageComponent	app-admin-management-page	Admin Feature	None	None	Frontend/src/app/features/admin/admin-management/pages/admin-management-page.component.ts
AssignFeeModalComponent	app-assign-fee-modal	Admin Feature	faculties, feeTypes, isOpen, students	close, saved, triggerAlert, triggerConfirm	Frontend/src/app/features/admin/billing-management/components/assign-fee-modal/assign-fee-modal.component.ts
AssignRoomModalComponent	app-assign-room-modal	Admin Feature	application, hostels, isOpen	assigned, close	Frontend/src/app/features/admin/hostel-management/components/assign-room-modal/assign-room-modal.component.ts
AuditDiffModalComponent	app-audit-diff-modal	Admin Feature	allLogs, isOpen, log	close	Frontend/src/app/features/admin/audit-logs/pages/components/audit-diff-modal.component.ts
AuditLogsPageComponent	app-audit-logs-page	Admin Feature	None	None	Frontend/src/app/features/admin/audit-logs/pages/audit-logs-page.component.ts

Component Class	Selector	Category	Inputs	Outputs	Relative File Path
BillingDashboard Component	app-billing-dashboard	Admin Feature	None	None	Frontend/src/app/features/admin/billing-management/pages/billing-dashboard/billing-dashboard.component.ts
CertificateManagementPageComponent	app-certificate-management-page	Admin Feature	None	None	Frontend/src/app/features/admin/certificate-management/pages/certificate-management-page.component.ts
ComplaintManagementPageComponent	app-complaint-management-page	Admin Feature	None	None	Frontend/src/app/features/admin/complaint-management/pages/complaint-management-page.component.ts
CreateFeeTypeModalComponent	app-create-fee-type-modal	Admin Feature	isOpen	close, saved, triggerConfirm	Frontend/src/app/features/admin/billing-management/components/create-fee-type-modal/create-fee-type-modal.component.ts
CreateRoomModalComponent	app-create-room-modal	Admin Feature	hostels, initialHostelId, isOpen, roomToEdit	close, saved	Frontend/src/app/features/admin/hostel-management/components/create-room-modal/create-room-modal.component.ts
EventManagerPageComponent	app-event-management-page	Admin Feature	None	None	Frontend/src/app/features/admin/event-management/pages/event-management-page.component.ts
FacultyManagementPageComponent	app-faculty-management-page	Admin Feature	None	None	Frontend/src/app/features/admin/faculty-management/pages/faculty-management.page/faculty-management.page.component.ts
FeeAssignmentsListComponent	app-fee-assignments-list	Admin Feature	currentPage, isLoading, ledgerItems, pageSize, totalRecords	cancelUnpaid, pageChange, pageSizeChange, refresh	Frontend/src/app/features/admin/billing-management/components/fee-assignments-list/fee-assignments-list.component.ts

COMPONENT CLASS	SELECTOR	CATEGORY	INPUTS	OUTPUTS	RELATIVE FILE PATH
FeeTypesListComponent	app-fee-types-list	Admin Feature	feeTypes , isLoading	deactivate , refresh , toggleStatus	Frontend/src/app/features/admin/billing-management/components/fee-types-list/fee-types-list.component.ts
HostelManagementPageComponent	app-hostel-management-page	Admin Feature	None	None	Frontend/src/app/features/admin/hostel-management/pages/hostel-management-page.component.ts
HostelsRoomsDirectoryComponent	app-hostels-rooms-directory	Admin Feature	isLoading , set	createRoom , editRoom , hostelTabChange , refreshed , triggerConfirm	Frontend/src/app/features/admin/hostel-management/components/hostels-rooms-directory/hostels-rooms-directory.component.ts
LabBookingsHistoryComponent	app-lab-bookings-history	Admin Feature	bookings , loading , pageSize	None	Frontend/src/app/features/admin/lab-management/components/lab-bookings-history/lab-bookings-history.component.ts
LabDetailsComponent	app-lab-details	Admin Feature	isModal , lab	closeModal , createLabSubmitted , inspectLayout	Frontend/src/app/features/admin/lab-management/components/lab-details/lab-details.component.ts
LabManagementPageComponent	app-lab-management-page	Admin Feature	None	None	Frontend/src/app/features/admin/lab-management/pages/lab-management-page.component.ts
LabTimeSlotsManagementComponent	app-lab-time-slots-management	Admin Feature	labs	None	Frontend/src/app/features/admin/lab-management/components/lab-time-slots-management/lab-time-slots-management.component.ts
NotificationMonitorPageComponent	app-notification-monitor-page.component	Admin Feature	None	None	Frontend/src/app/features/admin/notification-monitor/pages/notification-monitor-page.component.component.ts

COMPONENT CLASS	SELECTOR	CATEGORY	INPUTS	OUTPUTS	RELATIVE FILE PATH
ReportsDashboardComponent	app-reports-dashboard	Admin Feature	None	None	Frontend/src/app/features/admin/reports/pages/reports-dashboard/reports-dashboard.component.ts
StudentMasterPageComponent	app-student-master-page	Admin Feature	None	None	Frontend/src/app/features/admin/student-master/pages/student-master-page.component.ts
SystemSettingsPageComponent	app-system-settings-page	Admin Feature	None	None	Frontend/src/app/features/admin/system-settings/pages/system-settings-page.component.ts
UpdateTicketModalComponent	app-update-ticket-modal	Admin Feature	None	None	Frontend/src/app/features/admin/complaint-management/components/update-ticket-modal/update-ticket-modal.component.ts
ForgotPasswordComponent	app-forgot-password	Auth Feature	None	None	Frontend/src/app/features/auth/forgot-password/forgot-password.component.ts
LoginComponent	app-login	Auth Feature	None	None	Frontend/src/app/features/auth/login/login.component.ts
ResetPasswordPageComponent	app-reset-password.page	Auth Feature	None	None	Frontend/src/app/features/auth/reset-password/pages/reset-password.page/reset-password.page.component.ts
StudentRegistrationComponent	app-student-registration	Auth Feature	None	None	Frontend/src/app/features/auth/student-registration/student-registration.component.ts
VerifyEmailPageComponent	app-verify-email.page	Auth Feature	None	None	Frontend/src/app/features/auth/verify-email/pages/verify-email.page/verify-email.page.component.ts
AuthLayoutComponent	app-auth-layout	Layout	None	None	Frontend/src/app/layouts/auth-layout/auth-layout.component.ts

Component Class	Selector	Category	Inputs	Outputs	Relative File Path
MainLayoutComponent	app-main-layout	Layout	None	None	Frontend/src/app/layouts/main-layout/main-layout.component.ts
ActionButtonComponent	app-action-button	Shared	disabled, fullWidth, icon, iconPosition, imageSrc, label, loading, size, tooltip, tooltipPosition, type, variant	btnClick	Frontend/src/app/shared/components/action-button/action-button.component.ts
AddressEditorComponent	app-address-editor	Shared	addressForm	None	Frontend/src/app/shared/components/forms/address-editor/address-editor.component.ts
AlertModalComponent	app-alert-modal	Shared	buttonIcon, buttonText, buttonVariant, icon, iconVariant, isOpen, message, title	close	Frontend/src/app/shared/components/dialogs/alert-modal/alert-modal.component.ts
AmountCardComponent	app-amount-card	Shared	amount, icon, isLoading, variant	None	Frontend/src/app/shared/components/cards/amount-card/amount-card.component.ts
AnalyticsChartComponent	app-analytics-chart	Shared	None	None	Frontend/src/app/shared/components/analytics-chart/chart/analytics-chart.component.ts
ApplicationsTableViewComponent	app-applications-table-view	Shared	None	None	Frontend/src/app/features/hostel-shared/components/applications-table-view/applications-table-view.component.ts
BookingSelectorsComponent	app-booking-selectors	Shared	allowPastDates, isLoading, labs, selectedDate, selectedLabId, selectedTimeSlot, timeSlotsInput	dateChange, labChange, refresh, timeSlotChange	Frontend/src/app/features/lab-shared/components/booking-selectors/booking-selectors.component.ts
ConfirmModalComponent	app-confirm-modal	Shared	cancelIcon, cancelText, confirmIcon, confirmText, confirmVariant, icon, iconVariant, isConfirmDisabled, isLoading, isOpen, message, title	cancel, confirm	Frontend/src/app/shared/components/dialogs/confirm-modal/confirm-modal.component.ts
ContentPanelComponent	app-content-panel	Shared	None	None	Frontend/src/app/shared/components/content-panel/content-panel.component.ts

COMPONENT CLASS	SELECTOR	CATEGORY	INPUTS	OUTPUTS	RELATIVE FILE PATH
ControlPanelCard Component	app-control-panel-card	Shared	None	cardClick	Frontend/src/app/shared/components/cards/control-panel-card/control-panel-card.component.ts
DashboardCardComponent	app-dashboard-card	Shared	icon, iconVariant, isLoading, linkRoute, linkText, value, valueVariant	None	Frontend/src/app/shared/components/cards/dashboard-card/dashboard-card.component.ts
DataTableComponent	app-data-table	Shared	columns, currentPage, data, emptyMessage, isLoading, pageSize, searchPlaceholder, serverSide, showSearch, sortColumn, sortDirection, sortIconStyle, totalRecords	filterChange, pageChange, pageSizeChange, sortChange	Frontend/src/app/shared/components/data-table/data-table.component.ts
DataTableComponent	app-data-table	Shared	columns, currentPage, data, emptyMessage, hoverable, isLoading, pageSize, pageSizeOptions, showPagination, striped, totalRecords	clearAllFiltersEvent, filterChange, pageChange, pageSizeChange, sortChange	Frontend/src/app/shared/components/tables-utilities/data-table/data-table.component.ts
DatePickerComponent	app-date-picker	Shared	disabled, label, minDate, selectedDate	dateChange	Frontend/src/app/shared/components/date-picker/date-picker.component.ts
DropdownSelect Component	app-dropdown-select	Shared	None	None	Frontend/src/app/shared/components/dropdown-select/dropdown-select.component.ts
EmptyStateComponent	app-empty-state	Shared	actionIcon, actionText, description, icon, title	actionClick	Frontend/src/app/shared/components/empty-state/empty-state.component.ts
ErrorStateComponent	app-error-state	Shared	message, retryLabel, title	retry	Frontend/src/app/shared/components/error-state/error-state.component.ts
EventCardComponent	app-event-card	Shared	isProcessing	cancel, register	Frontend/src/app/shared/components/cards/event-card/event-card.component.ts
FilterComponent	app-filter	Shared	label, options, searchPlaceholder, selectedValues, showSearch	clear, filterChange	Frontend/src/app/shared/components/tables-utilities/filter/filter.component.ts

COMPONENT CLASS	SELECTOR	CATEGORY	INPUTS	OUTPUTS	RELATIVE FILE PATH
FooterComponent	app-footer-component	Shared	None	None	Frontend/src/app/shared/layout-components/footer-component/footer-component.component.ts
FormInputComponent	app-form-input	Shared	None	None	Frontend/src/app/shared/components/form-input/form-input.component.ts
HeaderComponent	app-header-component	Shared	None	None	Frontend/src/app/shared/layout-components/header-component/header-component.component.ts
LabGridMatrixComponent	app-lab-grid-matrix	Shared	isModal, isStudent, labId, labName, seats, showBookingStatus, showZoom, totalCols, totalRows	closeModal, seatAdded, seatRemoved, seatSelected	Frontend/src/app/features/lab-shared/components/lab-grid-matrix/lab-grid-matrix.component.ts
LoadingSpinnerComponent	app-loading-spinner	Shared	color, containerClasses, label, size	None	Frontend/src/app/shared/components/loading-spinner/loading-spinner.component.ts
LoadingStateComponent	app-loading-state	Shared	description, spinnerSize, title	None	Frontend/src/app/shared/components/loading-state/loading-state.component.ts
MasterLedgerTableComponent	app-master-ledger-table	Shared	None	None	Frontend/src/app/shared/components/master-ledger-table/master-ledger-table.component.ts
MultiSelectDropDownComponent	app-multi-select-dropdown	Shared	disabled, label, options, placeholder, showSearch	selectionChange	Frontend/src/app/shared/components/multi-select-dropdown/multi-select-dropdown.component.ts
NavbarComponent	app-navbar-component	Shared	None	None	Frontend/src/app/shared/layout-components/navbar-component/navbar-component.component.ts
NotFoundPageComponent	app-not-found-page	Shared	None	None	Frontend/src/app/shared/pages/not-found-page/not-found-page.component.ts

Component Class	Selector	Category	Inputs	Outputs	Relative File Path
NotificationItem Component	app-notification-item	Shared	None	None	Frontend/src/app/shared/components/notification-item/notification-item.component.ts
OtpVerificationModalComponent	app-otp-verification-modal	Shared	canClose , currentStep , emailAddress , isOpen , isResending , isVerifying , mode , phoneNumber , validityMinutes	cancelled , resendEmail , resendSms , verifyEmail , verifyOtp	Frontend/src/app/shared/components/modals/otp-verification-modal/otp-verification-modal.component.ts
PageHeaderComponent	app-page-header	Shared	badgeText , description , icon , showIndexBadge , title	None	Frontend/src/app/shared/components/page-header/page-header.component.ts
PaginationComponent	app-pagination	Shared	currentPage , pageSize , pageSizeOptions , showInfo , showPageSizeOptions , totalRecords	pageChange , pageSizeChange	Frontend/src/app/shared/components/tables-utilities/pagination/pagination.component.ts
PasswordChangeComponent	app-password-change	Shared	email , mode , resetToken , showCancelButton	cancel , success	Frontend/src/app/shared/components/password-change/password-change.component.ts
PhoneNumbersFormComponent	app-phone-numbers-form	Shared	isPrimaryVerified , phoneNumbersArray , readonly	None	Frontend/src/app/shared/components/forms/phone-numbers-form/phone-numbers-form.component.ts
ReportDetailGridComponent	app-report-detail-grid	Shared	columns , drilldownButtonLabel , emptyMessage , items , showDrilldown , showRowNumbers , sortBy , sortDirection , startIndex	drilldown , sort	Frontend/src/app/shared/components/reports/report-detail-grid/report-detail-grid.component.ts
ReportFooterComponent	app-report-footer	Shared	grandTotals , institutionName , preparedBy , printedDateTime , showSignOff , systemInfo	None	Frontend/src/app/shared/components/reports/report-footer/report-footer.component.ts
ReportGroupFooterComponent	app-report-group-footer	Shared	label , subtotals	None	Frontend/src/app/shared/components/reports/report-group-footer/report-group-footer.component.ts
ReportGroupHeaderComponent	app-report-group-header	Shared	badge , count , isExpanded , submetrics , title	toggle	Frontend/src/app/shared/components/reports/report-group-header/report-group-header.component.ts

Component Class	Selector	Category	Inputs	Outputs	Relative File Path
ReportHeaderComponent	app-report-header	Shared	academicTerm , filterChips , generatedBy , generatedDate , institutionName , subtitle , title	None	Frontend/src/app/shared/components/reports/report-header/report-header.component.ts
ReportPageSetupComponent	app-report-page-setup	Shared	isFullscreen , orientation , paperSize , zoom	exportCsv , exportExcel , exportPdf , exportWord , orientationChange , paperSizeChange , print , toggleFullscreen , zoomChange	Frontend/src/app/shared/components/reports/report-page-setup/report-page-setup.component.ts
ReportPaginationComponent	app-report-pagination	Shared	hasNextPage , hasPreviousPage , pageNumber , pageSize , totalCount , totalPages	pageChange , pageSizeChange	Frontend/src/app/shared/components/reports/report-pagination/report-pagination.component.ts
ReportSubreportComponent	app-report-subreport	Shared	columns , isOpen , items , loading , parentCategory , title	close	Frontend/src/app/shared/components/reports/report-subreport/report-subreport.component.ts
ReportViewerComponent	app-report-viewer	Shared	documentTitle , exportColumns , exportData , grandTotals , isFullscreen , orientation , paperSize , showToolbar , zoom	exportCsvTriggered , exportExcelTriggered , exportPdfTriggered , exportWordTriggered , fullscreenChange	Frontend/src/app/shared/components/reports/report-viewer/report-viewer.component.ts
SearchComponent	app-search	Shared	debounceMs , placeholder , value	clear , searchChange	Frontend/src/app/shared/components/tables-utilities/search/search.component.ts
SelectDropdownComponent	app-select-dropdown	Shared	icon , label , options , placeholder , required , searchable , selectedValue	selectionChange	Frontend/src/app/shared/components/select-dropdown/select-dropdown.component.ts
SortComponent	app-sort	Shared	activeColumn , column , direction , label	sortChange	Frontend/src/app/shared/components/tables-utilities/sort/sort.component.ts
StatusBadgeComponent	app-status-badge	Shared	count , countUnit , icon , pulse , size , status , variant	None	Frontend/src/app/shared/components/status-badge/status-badge.component.ts

COMPONENT CLASS	SELECTOR	CATEGORY	INPUTS	OUTPUTS	RELATIVE FILE PATH
TabComponent	app-tab-component	Shared	activeTabId , tabs , variant	tabChange	Frontend/src/app/shared/components/tab-component/tab.component.ts
TimePickerComponent	app-time-picker	Shared	disabled , icon , label , minuteStep , selectedTime , showLabel	timeChange	Frontend/src/app/shared/components/time-picker/time-picker.component.ts
ToastContainerComponent	app-toast-container	Shared	None	None	Frontend/src/app/shared/components/toast-container/toast-container.component.ts
ToastNotificationComponent	app-toast-notification	Shared	None	None	Frontend/src/app/shared/components/dialogs/toast-notification/toast-notification.component.ts
UnauthorizedPageComponent	app-unauthorized-page	Shared	None	None	Frontend/src/app/shared/pages/unauthorized-page/unauthorized-page.component.ts
ActiveHoldTimerComponent	app-active-hold-timer	Student Feature	activeBooking , holdMinutes , isSubmitting	confirm , holdExpired , releaseClick	Frontend/src/app/features/student/lab-booking/components/active-hold-timer/active-hold-timer.component.ts
ApplicationFormComponent	app-application-form	Student Feature	None	None	Frontend/src/app/features/student/hostel/hostel-application-page/components/application-form/application-form.component.ts
ApplicationFormComponent	app-application-form	Student Feature	None	None	Frontend/src/app/features/student/hostel-application/pages/components/application-form/application-form.component.ts
BillLedgerListComponent	app-bill-ledger-list	Student Feature	isLoading , ledgerItems	payOnline , refresh , viewReceipt	Frontend/src/app/features/student/billing/components/bill-ledger-list/bill-ledger-list.component.ts

COMPONENT CLASS	SELECTOR	CATEGORY	INPUTS	OUTPUTS	RELATIVE FILE PATH
CertificateRequestPageComponent	app-certificate-request-page	Student Feature	None	None	Frontend/src/app/features/student/certificate/pages/certificate-request-page.component.ts
HostelApplicationPageComponent	app-hostel-application-page	Student Feature	None	None	Frontend/src/app/features/student/hostel-application/pages/hostel-application-page.component.ts
LabBookingPageComponent	app-lab-booking-page	Student Feature	None	None	Frontend/src/app/features/student/lab-booking/pages/lab-booking-page.component.ts
PaymentCheckoutComponent	app-payment-checkout	Student Feature	isSubmitting, item	cancelCheckout, submitPayment	Frontend/src/app/features/student/billing/components/payment-checkout/payment-checkout.component.ts
StudentBillingPageComponent	app-student-billing-page	Student Feature	None	None	Frontend/src/app/features/student/billing/pages/student-billing-page.component.ts
StudentComplaintPageComponent	app-student-complaint-page	Student Feature	None	None	Frontend/src/app/features/student/complaint/pages/student-complaint-page.component.ts
StudentDashboardPageComponent	app-student-dashboard-page	Student Feature	None	None	Frontend/src/app/features/student/dashboard/pages/student-dashboard-page.component.ts
StudentEventPageComponent	app-student-event-page	Student Feature	None	None	Frontend/src/app/features/student/event/pages/student-event-page.component.ts
StudentHistoryListComponent	app-student-history-list	Student Feature	bookings, isLoading	cancelBooking, refresh	Frontend/src/app/features/student/lab-booking/components/student-history-list/student-history-list.component.ts
StudentNotificationPageComponent	app-student-notification-page	Student Feature	None	None	Frontend/src/app/features/student/notification/pages/student-notification-page/student-notification-page.component.ts

COMPONENT CLASS	SELECTOR	CATEGORY	INPUTS	OUTPUTS	RELATIVE FILE PATH
StudentPaymentPageComponent	app-student-payment-page	Student Feature	None	None	Frontend/src/app/features/student/billing/pages/student-payment-page.component.ts
StudentSecurityPageComponent	app-student-security-page	Student Feature	None	None	Frontend/src/app/features/student/security/pages/student-security-page.component.ts
StudentSettingsPageComponent	app-student-settings-page	Student Feature	None	None	Frontend/src/app/features/student/settings/pages/student-settings-page.component.ts
booking-selectors	N/A	Student Feature	None	None	Frontend/src/app/features/student/lab-booking/components/booking-selectors/booking-selectors.component.ts

Part 2: Routing Table

The portal implements lazy loading via `loadChildren` and `loadComponent` across 20 routing modules. The root routing table configures feature shells, authentication guards, and terminal wildcard fallbacks.

ROUTE PATH	TYPE	GUARD(S)	TARGET COMPONENT / CHUNK	DESCRIPTION
/	Redirect	-	redirectTo: '/login'	Root redirection to primary authentication page
/login	Public	-	LoginComponent	Student & Administrator sign-in portal
/register	Public	-	RegisterComponent	Student self-service registration with index validation
/forgot-password	Public	-	ForgotPasswordComponent	Password recovery initiation request
/reset-password	Public	-	ResetPasswordComponent	Tokenized password reset form
/admin	Shell	AuthGuard , AdminGuard	AdminLayoutComponent	Admin master shell hosting nested <router-outlet>
/admin/dashboard	Feature	AuthGuard , AdminGuard	AdminDashboardPageComponent	Real-time KPIs, quick actions, and pending triage overview
/admin/student-master	Feature	AuthGuard , AdminGuard	StudentMasterComponent	Student directory, intake CSV bulk import, and account status
/admin/faculties	Feature	AuthGuard , AdminGuard	FacultyManagementPageComponent	Faculty creation, department assignment, and cohort sizes
/admin/hostels	Feature	AuthGuard , AdminGuard	AdminHostelPageComponent	Hostel blocks, room capacity grids, and allocation approvals

ROUTE PATH	TYPE	GUARD(S)	TARGET COMPONENT / CHUNK	DESCRIPTION
/admin/labs	Feature	AuthGuard , AdminGuard	LabManagementPageComponent	2D Computer workstation builder, science lab batches, zoom
/admin/events	Feature	AuthGuard , AdminGuard	EventManagerPageComponent	Campus event scheduling, venue bookings, and attendee rosters
/admin/complaints	Feature	AuthGuard , AdminGuard	ComplaintManagementPageComponent	Ticket triage, priority elevation, and administrative replies
/admin/certificates	Feature	AuthGuard , AdminGuard	CertificateManagementPageComponent	Official Bonafide, Transcript, and Completion letter processing
/admin/billing	Feature	AuthGuard , AdminGuard	BillingDashboardComponent	Tuition fee types, single student assignment, faculty bulk runs
/admin/notifications	Feature	AuthGuard , AdminGuard	NotificationMonitorPageComponent	Broadcast announcements and automated notification audit
/admin/analytics	Feature	AuthGuard , AdminGuard	AdminReportsPageComponent	Institutional metrics, visual charts, and exportable summaries
/admin/audit-logs	Feature	AuthGuard , AdminGuard	AuditLogsPageComponent	Immutable audit trail, JSON diff viewer, user activity search
/admin/user-management	Feature	AuthGuard , AdminGuard	AdminManagementPageComponent	Administrator credentials, role elevation, and deactivation
/admin/system-settings	Feature	AuthGuard , AdminGuard	SystemSettingsPageComponent	Global pagination, semester deadlines, and maintenance flags
/student	Shell	AuthGuard	StudentLayoutComponent	Student portal master layout with personal navigation
/student/dashboard	Feature	AuthGuard	StudentDashboardPageComponent	Personalized student summary, active allocations, announcements
/student/profile	Feature	AuthGuard	StudentProfilePageComponent	Student details, verified contacts, and password management
/student/hostels	Feature	AuthGuard	HostelBookingPageComponent	Hostel room inspection and lease application workflow
/student/lab-booking	Feature	AuthGuard	LabBookingPageComponent	Interactive 2D workstation seat reservation with active timer
/student/events	Feature	AuthGuard	StudentEventPageComponent	University events catalog and registration ticket passes
/student/complaints	Feature	AuthGuard	StudentComplaintPageComponent	Ticket lodging with image attachments and real-time status
/student/certificates	Feature	AuthGuard	StudentCertificatePageComponent	Request letter generation with purpose specification
/student/billing	Feature	AuthGuard	StudentPaymentPageComponent	Ledger breakdown, due fee inspection, and receipt download
/student/notifications	Feature	AuthGuard	StudentNotificationPageComponent	System messages, unread badge counter, and mark-read actions
**	Wildcard	-	NotFoundComponent	Terminal 404 page with return-to-safety routing

Part 3: Service Inventory (27 Services)

All services are provided in the application root (`providedIn: 'root'`) and manage typed asynchronous communication with the ASP.NET Core Web API.

SERVICE NAME	CATEGORY	MANAGED API ENDPOINTS	KEY METHODS	RELATIVE FILE PATH
AdminDashboardService	Admin	Dynamic API Mapping	<code>getAdminDashboardMetrics()</code> , <code>getAllSystemSettings()</code> , <code>getDefaultPageSize()</code> , <code>getHoldMinutes()</code> , <code>saveDefaultPageSize()</code> , <code>saveHoldMinutes()</code>	<code>Frontend/src/app/features/admin/dashboard/services/admin-dashboard.service.ts</code>
CertificateManagementService	Admin	Dynamic API Mapping	<code>createCertificateType()</code> , <code>deleteCertificateType()</code> , <code>getCertificateTypes()</code> , <code>getRequests()</code> , <code>updateCertificateType()</code> , <code>updateRequestStatus()</code>	<code>Frontend/src/app/features/admin/certificate-management/services/certificate-management.service.ts</code>
ComplaintManagementService	Admin	Dynamic API Mapping	<code>createCategory()</code> , <code>deleteCategory()</code> , <code>getCategories()</code> , <code>getComplaints()</code> , <code>updateCategory()</code> , <code>updateComplaintStatus()</code>	<code>Frontend/src/app/features/admin/complaint-management/services/complaint-management.service.ts</code>
EventManagerService	Admin	Dynamic API Mapping	<code>createEvent()</code> , <code>createVenue()</code> , <code>deleteVenue()</code> , <code>getEventRegistrations()</code> , <code>getEvents()</code> , <code>getFormattedEventRegistrations()</code> (+4 more)	<code>Frontend/src/app/features/admin/event-management/services/event-management.service.ts</code>
FacultyManagementService	Admin	Dynamic API Mapping	<code>createFaculty()</code> , <code>deleteFaculty()</code> , <code>getFaculties()</code> , <code>getFormattedFaculties()</code> , <code>updateFaculty()</code>	<code>Frontend/src/app/features/admin/faculty-management/services/faculty-management.service.ts</code>
HostelManagementService	Admin	Dynamic API Mapping	<code>assignRoom()</code> , <code>createHostel()</code> , <code>createRoom()</code> , <code>deleteHostel()</code> , <code>deleteRoom()</code> , <code>getAllApplications()</code> (+7 more)	<code>Frontend/src/app/features/admin/hostel-management/services/hostel-management.service.ts</code>
LabManagementService	Admin	Dynamic API Mapping	<code>addSeat()</code> , <code>createLab()</code> , <code>createTimeSlot()</code> , <code>deleteTimeSlot()</code> , <code>getAvailableTimeSlots()</code> , <code>getBookingsHistory()</code> (+7 more)	<code>Frontend/src/app/features/admin/lab-management/services/lab-management.service.ts</code>
NotificationMonitorService	Admin	Dynamic API Mapping	<code>getAdminAuditLog()</code> , <code>getFaculties()</code> , <code>getStudentsDirectory()</code> , <code>markAsRead()</code> , <code>sendInternalNotification()</code>	<code>Frontend/src/app/features/admin/notification-monitor/services/notification-monitor.service.ts</code>
StudentMasterService	Admin	Dynamic API Mapping	<code>checkDeactivateSafety()</code> , <code>deactivateAccount()</code> , <code>loadAllMasterRecords()</code> , <code>loadFaculties()</code> , <code>loadMasterList()</code> , <code>loadStudentAccounts()</code> (+3 more)	<code>Frontend/src/app/features/admin/student-master/services/student-master.service.ts</code>
system-settings.service.ts	Admin	Dynamic API Mapping	<i>Signals / Helper methods</i>	<code>Frontend/src/app/features/admin/system-settings/services/system-settings.service.ts</code>
ApiService	Core	Dynamic API Mapping	<i>Signals / Helper methods</i>	<code>Frontend/src/app/core/services/api.service.ts</code>
AuditLogService	Core	Dynamic API Mapping	<code>acknowledgeAll()</code> , <code>acknowledgeLog()</code> , <code>getAuditLogById()</code> , <code>getAuditLogs()</code>	<code>Frontend/src/app/core/services/audit-log.service.ts</code>

Service Name	Category	Managed API Endpoints	Key Methods	Relative File Path
AuthService	Core	Dynamic API Mapping	changePassword(), clearMustChangePassword(), getPasswordPolicy(), login(), logout(), refreshTokenSession() (+5 more)	Frontend/src/app/core/services/auth.service.ts
ReportExportService	Core	Dynamic API Mapping	downloadBlob(), exportToCsv(), exportToExcel(), exportToPdf(), exportToWord(), printReport()	Frontend/src/app/core/services/report-export.service.ts
ReportService	Core	Dynamic API Mapping	getBillingReport(), getCertificateReport(), getCertificateTypesCatalogReport(), getComplaintCategoriesSlaReport(), getComplaintReport(), getEventReport() (+11 more)	Frontend/src/app/core/services/report.service.ts
SystemSettingsService	Core	Dynamic API Mapping	getAllSettings(), getDefaultPageSize(), getHoldMinutes(), loadDefaultPageSize(), updateDefaultPageSize(), updateHoldMinutes() (+1 more)	Frontend/src/app/core/services/system-settings.service.ts
ThemeService	Core	Dynamic API Mapping	applyTheme(), setDarkMode(), setTheme(), toggleDarkMode()	Frontend/src/app/core/services/theme.service.ts
ToastService	Core	Dynamic API Mapping	clear(), remove()	Frontend/src/app/core/services/toast.service.ts
CertificateRequestService	Student	Dynamic API Mapping	getCertificateTypes(), getMyRequests(), submitRequest()	Frontend/src/app/features/student/certificate/services/certificate-request.service.ts
EventService	Student	Dynamic API Mapping	cancelRegistration(), getAvailableEvents(), getFormattedEvents(), registerForEvent()	Frontend/src/app/features/student/event/services/event.service.ts
HostelApplicationService	Student	Dynamic API Mapping	getFormattedHostelsLookup(), getHostelsLookup(), getMyApplications(), getMyFormattedApplications(), submitApplication()	Frontend/src/app/features/student/hostel-application/services/hostel-application.service.ts
LabBookingService	Student	Dynamic API Mapping	cancelBooking(), confirmBooking(), confirmHold(), createHold(), getAvailableTimeSlots(), getLabLayout() (+4 more)	Frontend/src/app/features/student/lab-booking/services/lab-booking.service.ts
StudentBillingService	Student	Dynamic API Mapping	getFormattedLedger(), getInvoiceDetails(), getStudentLedger(), payInvoice(), requestPaymentOtp()	Frontend/src/app/features/student/billing/services/student-billing.service.ts
StudentComplaintService	Student	Dynamic API Mapping	getCategories(), getMyComplaints(), submitComplaint()	Frontend/src/app/features/student/complaint/services/student-complaint.service.ts
StudentDashboardService	Student	Dynamic API Mapping	getDashboardMetrics(), getStudentProfile(), updateStudentProfile()	Frontend/src/app/features/student/dashboard/services/student-dashboard.service.ts
StudentNotificationService	Student	Dynamic API Mapping	getFormattedNotifications(), getMyNotifications(), markAsRead(), markAsUnread()	Frontend/src/app/features/student/notification/services/student-notification.service.ts
StudentSettingsService	Student	Dynamic API Mapping	selectTheme(), setFontSize(), setHeaderGradient(), toggleDarkMode()	Frontend/src/app/features/student/settings/services/student-settings.service.ts

Part 4: Forms Register

The frontend incorporates both **Reactive Forms** (for dynamic, multi-field, and mission-critical workflows) and **Template-driven Forms** (for standard authentication and simple settings).

FORM IDENTIFIER	FEATURE AREA	PARADIGM	STRUCTURE & KEY CONTROLS	VALIDATION RULES & ERROR HANDLING
Student Registration Form	Authentication	Reactive	Full Name, Index Number, University Email, Password, Confirm Password, Faculty Selection	Regex Index check (PS/YYYY/NNNN), university email domain check, cross-field password match validator. Errors displayed on touch.
User Sign-In Form	Authentication	Template-driven	Email / Username, Password, Remember Me checkbox	Required field validation, standard email formatting. Submit disabled while in-flight.
Profile & Contacts Editor	Student Profile	Reactive	Contact Details, FormArray of Phone Numbers (Primary, Mobile), FormArray of Addresses	FormArray dynamic row addition/deletion; phone number regex; mobile OTP verification requirement.
Hostel Application Form	Hostel	Reactive	Hostel Selection, Room Category, Academic Term Start Date, Academic Term End Date, Reason	Cross-field date validator ensuring EndDate > StartDate; room capacity check.
2D Workstation Builder Form	Lab Management	Reactive	Nested FormArray: Root FormGroup -> rows (FormArray) -> columns (FormArray) -> Workstation metadata	Automatic coordinate indexing (Row, Col) ; duplicate workstation identifier prevention.
Lab Seat Reservation Form	Lab Booking	Reactive	Lab Selection, Date Picker, Time Slot Dropdown, Workstation Selection	Custom future-date validator; slot availability check; 10-minute seat hold timer lock.
Complaint Lodging Form	Complaints	Reactive Dynamic	Category Dropdown, Title, Dynamic Description Fields, Urgency Level, Optional Image Upload	Dynamic control injection based on selected category; file size/type validation.
Certificate Request Form	Certificates	Reactive	Certificate Type Dropdown, Purpose Text, Delivery Mode, Copies Counter (1-5)	Number of copies minimum/maximum bounds (1-5); duplicate active request prevention.
Fee Assignment Modal Form	Billing	Reactive	Scope Mode (Single vs Faculty Bulk), Student Search, Faculty Cohort Dropdown, Fee Type, Amount, Due Date	Strict positive amount (> \$0.00); future due date; automatic student count check (disables bulk run if 0 enrolled).
System Settings Form	System Admin	Template-driven	Default Page Size, Maintenance Mode Toggle, Current Academic Year, Default Billing Term	Min/max bounds on pagination (5-100); immediate system effect signal updates.